CHEST X-RAY
MULTI LABEL - CLASES BINARIAS

Francisco Miralles Ferrer

Aplicaciones de Reconocimiento de Formas y Aprendizaje Automático

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4. Análisis de Resolución de Imágenes
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INTRODUCCIÓN

(Yang et al., 2021)

- Tarea de clasificación multietiqueta en radiográfias de tórax.
- Subconjunto derivado de NIH ChestX-ray14. (Wang et al., 2017)

TRAIN	VAL	TEST
78.468	11.219	22.433
70%	10%	20%

- Atelectasis
- Cardiomegalia
- Efusión (derrame pleural)
- Infiltración
- Masa
- Nódulo
- Neumonía
- Neumotórax
- Consolidación
- Edema
- Enfisema
- Fibrosis
- Engrosamiento pleural
- Hernia

SOTA

(Yang et al., 2023)
28x28 Images

Methods	ChestMNIST	
	AUC	ACC
ResNet-18 (28)[10]	0.768	0.947
ResNet-18 (224)[10]	0.773	0.947
ResNet-50 (28)[10]	0.769	0.947
ResNet-50 (224)[10]	0.773	0.948
auto-sklearn[11]	0.649	0.779
AutoKeras[12]	0.742	0.937
Google AutoML Vision	**0.778**	**0.948**

(Hossain et al., 2026)
224x224 Images

- AUC = 0.835 con Modelos Fundacionales en ChestMNIST.

(Strick et al., 2025)
224x224 Images NIHChestX-ray14

- AUC = 0.8527
- F1 = 0.3861

DannyNet (Densenet-121 pretrained-based)

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EDA

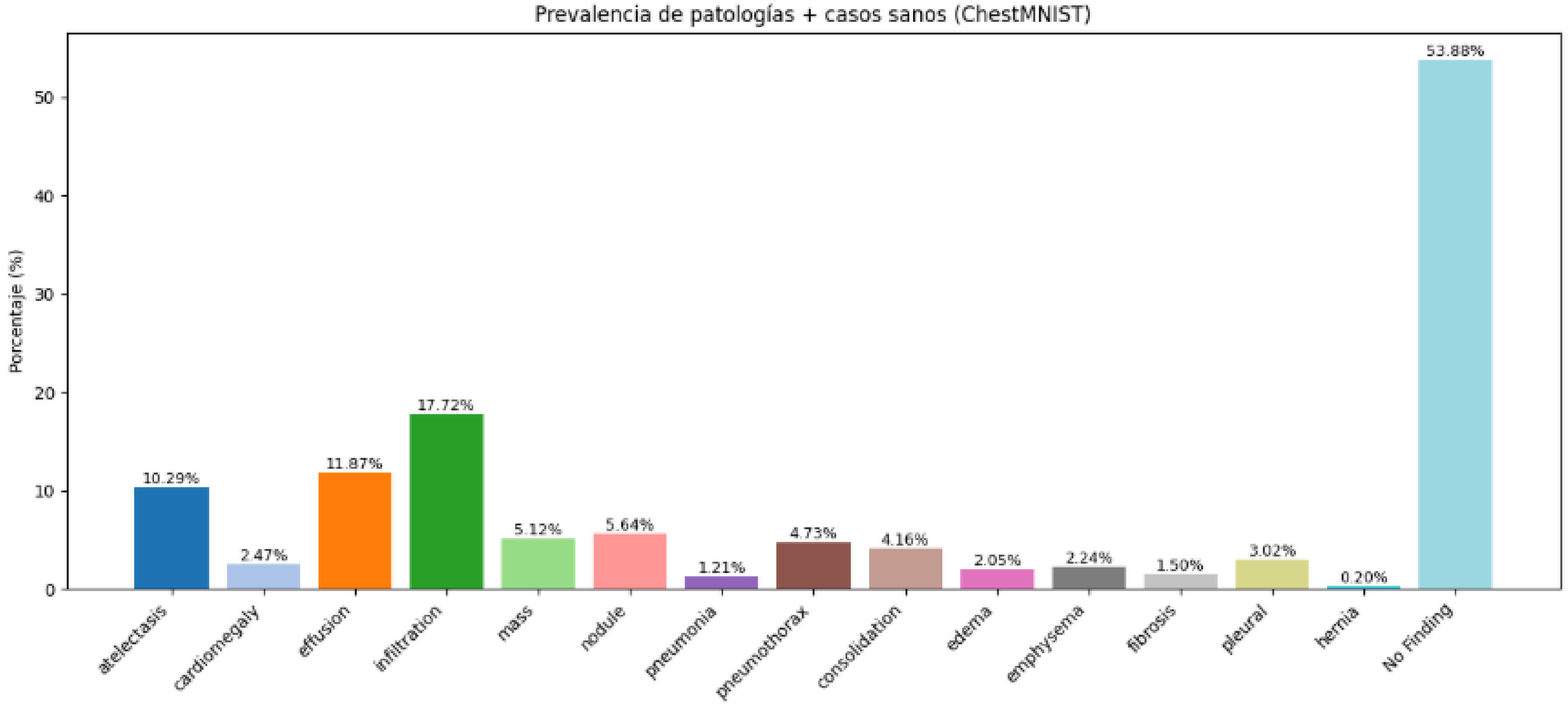
- 53'88% de los pacientes están sanos.
- 46'12% restante tienen almenos una patología.

- Muchos de los datos son ceros (predicción negativa).
- Modelos “Negacionistas” obtienen alta exactitud, pero carecen de interés médico.

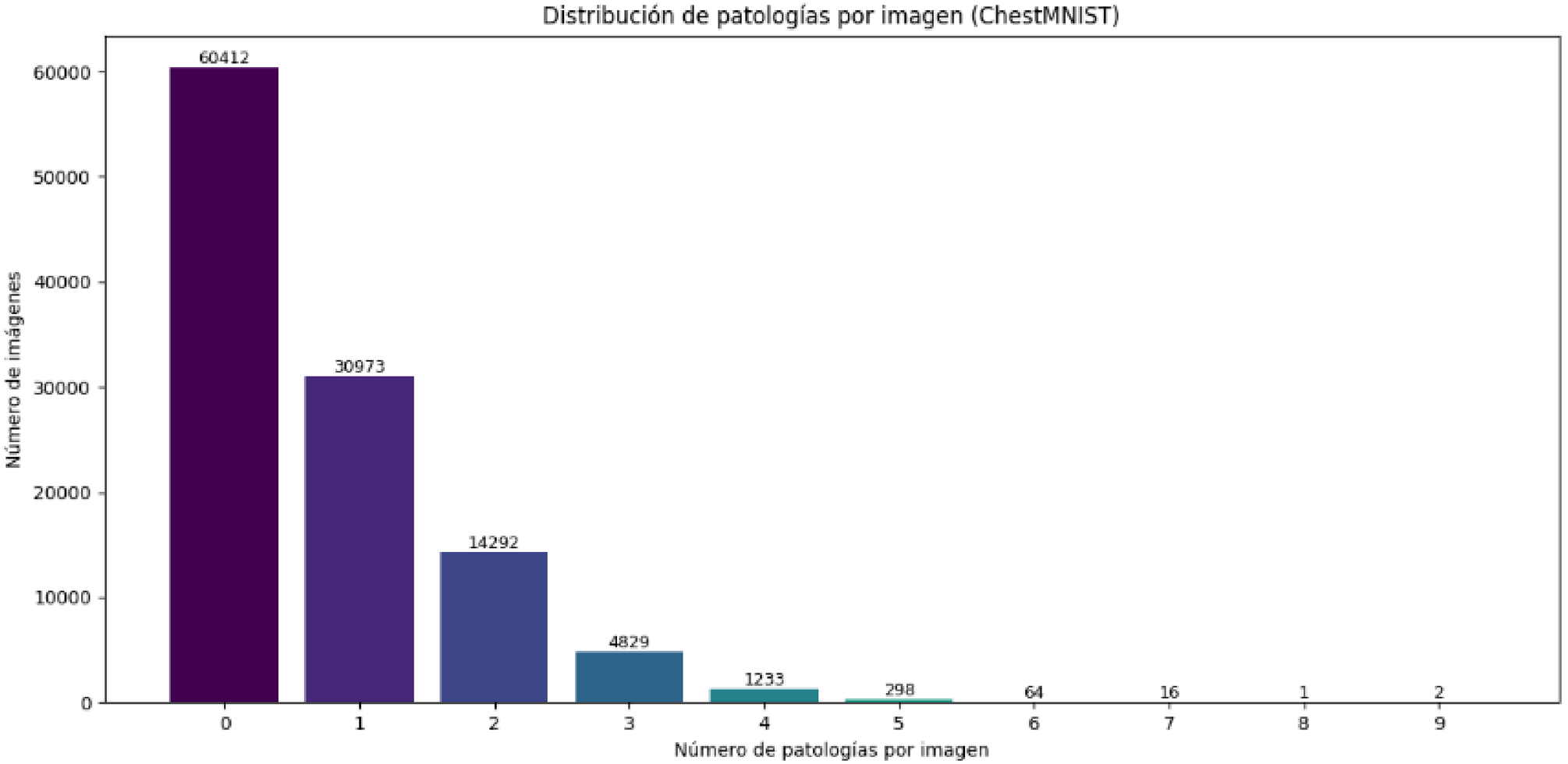
```
[0 0 0 0 0 0 0 0 0 0 0 0 0 0]
[0 0 0 0 0 0 0 0 0 0 0 0 0 0]
[0 0 0 0 0 0 0 0 0 0 0 0 0 0]
[0 0 0 0 0 0 0 0 0 0 0 0 0 0]
[0 0 0 0 0 0 0 0 0 0 0 0 0 0]
[0 0 0 1 0 1 0 0 0 0 0 0 0 0]
[0 0 0 1 0 0 0 1 1 0 1 0 0 0]
[0 0 0 0 0 0 0 1 0 0 0 0 0 0]
[0 0 1 1 0 0 0 0 0 0 0 0 0 0]
```

Atelectasis
cardiomegaly
Effusion
Infiltration
Mass
Nodule
Pneumonia
Pneumothorax
Consolidation
Edema
Emphysema
Fibrosis
Pleural
Hernia

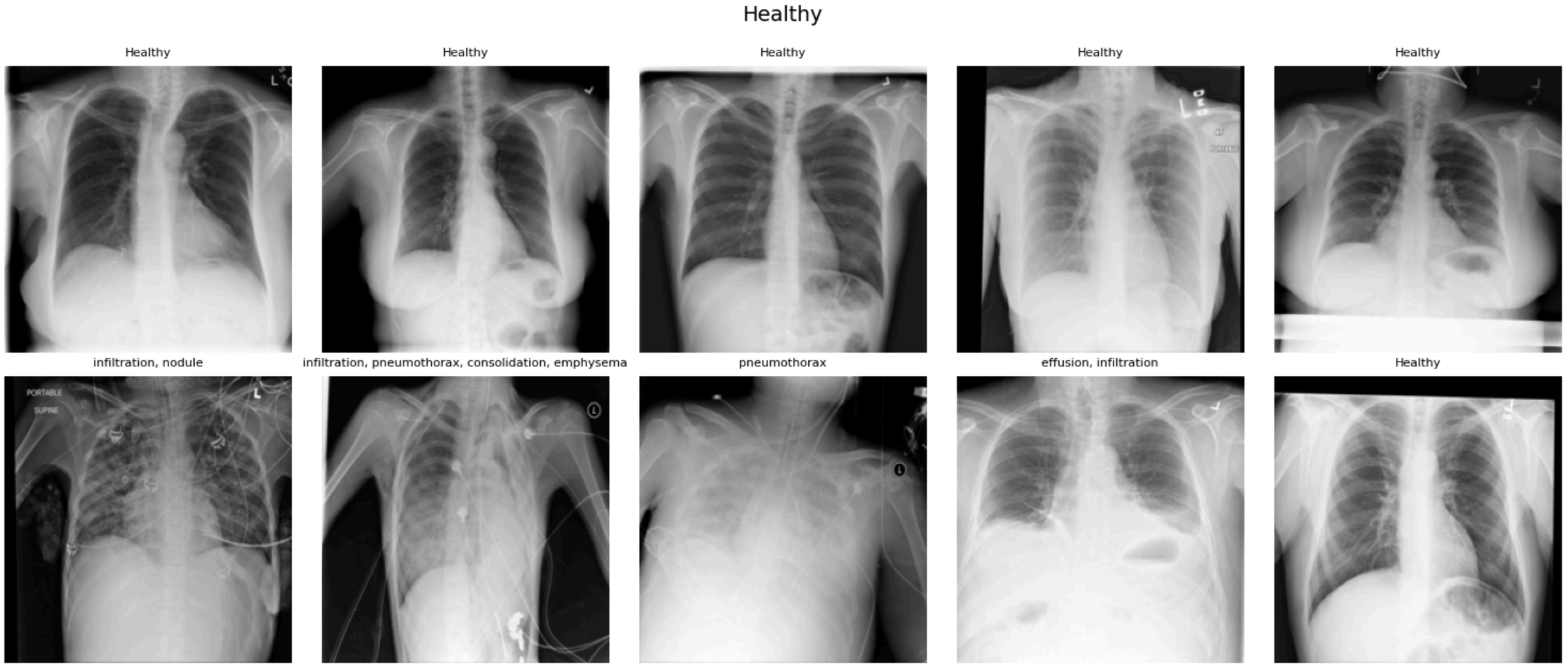
EDA


Prevalencia de patologías + casos sanos (ChestMNIST)
Porcentaje (%)
0
10
20
30
40
50
10.29%
2.47%
11.87%
17.72%
5.12%
5.64%
1.21%
4.73%
4.16%
2.05%
2.24%
1.50%
3.02%
0.20%
53.88%
atelectasis
cardiomegaly
effusion
infiltration
mass
nodule
pneumonia
pneumothorax
consolidation
edema
emphysema
fibrosis
pleural
hernia
No Finding

EDA


Distribución de patologías por imagen (ChestMNIST)
Número de imágenes
Número de patologías por imagen
60412
30973
14292
4829
1233
298
64
16
1
2
0
10000
20000
30000
40000
50000
60000
0
1
2
3
4
5
6
7
8
9

VISUALIZACIÓN DE DATOS



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MÉTRICAS

- **AUC-ROC Macro Media:** área bajo la curva ROC media para cada etiqueta.
- **Hamming Score:** porcentaje de etiquetas correctamente predichas.
- **Mean Balanced Accuracy:** rendimiento equilibrado entre positivas y negativas.
- **F1-score macro:** equilibrio entre precisión y recall con cada clase el mismo peso.
- **F1-score micro:** equilibrio entre precisión y recall globalmente.
- **Accuracy Global (EMR):** muestras clasificadas completamente bien.
- **Matriz de confusión:** separa las predicciones frente a las etiquetas reales.

ANÁLISIS DE RESOLUCIÓN DE IMÁGENES

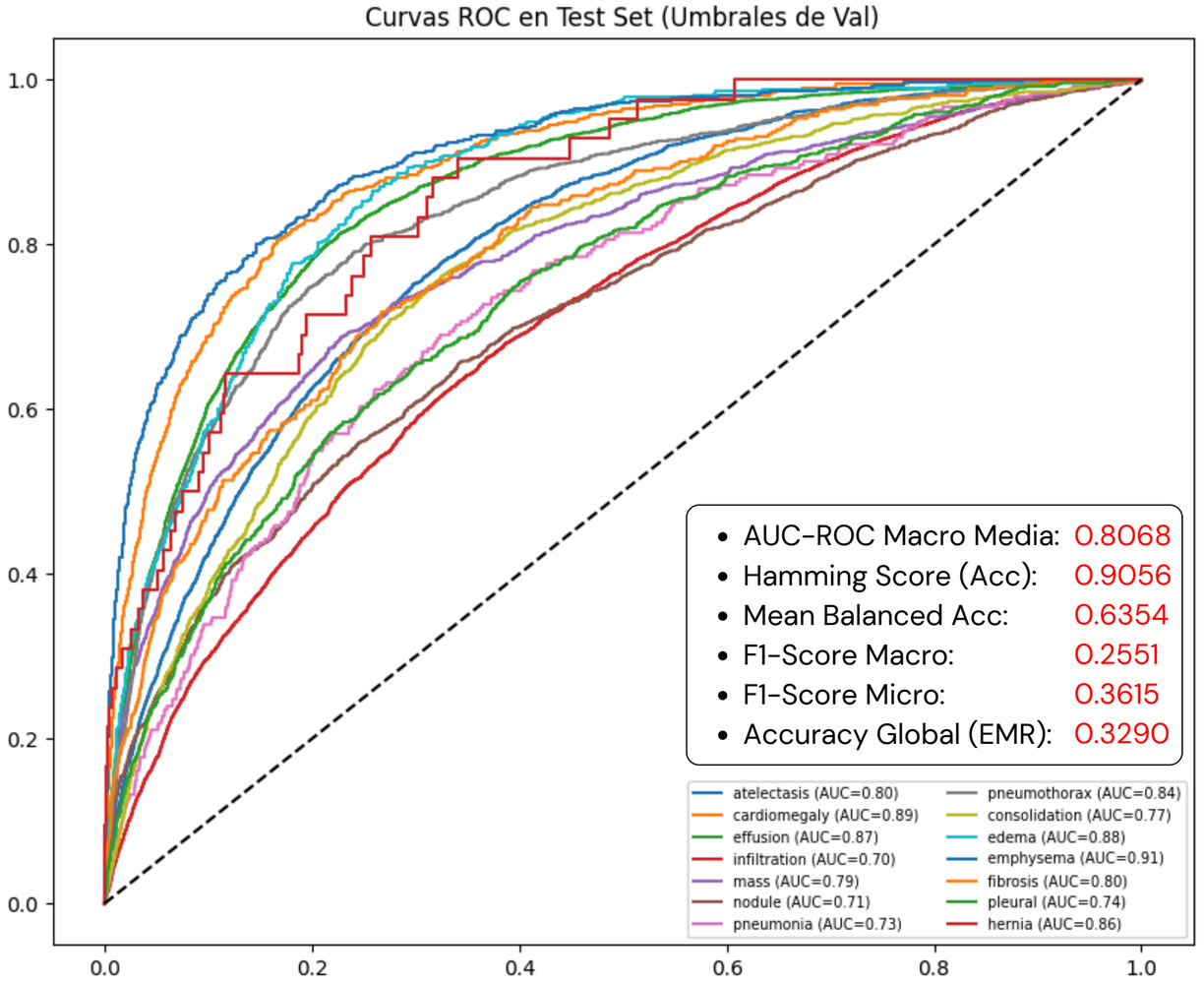
Rendimiento en diferentes Resoluciones

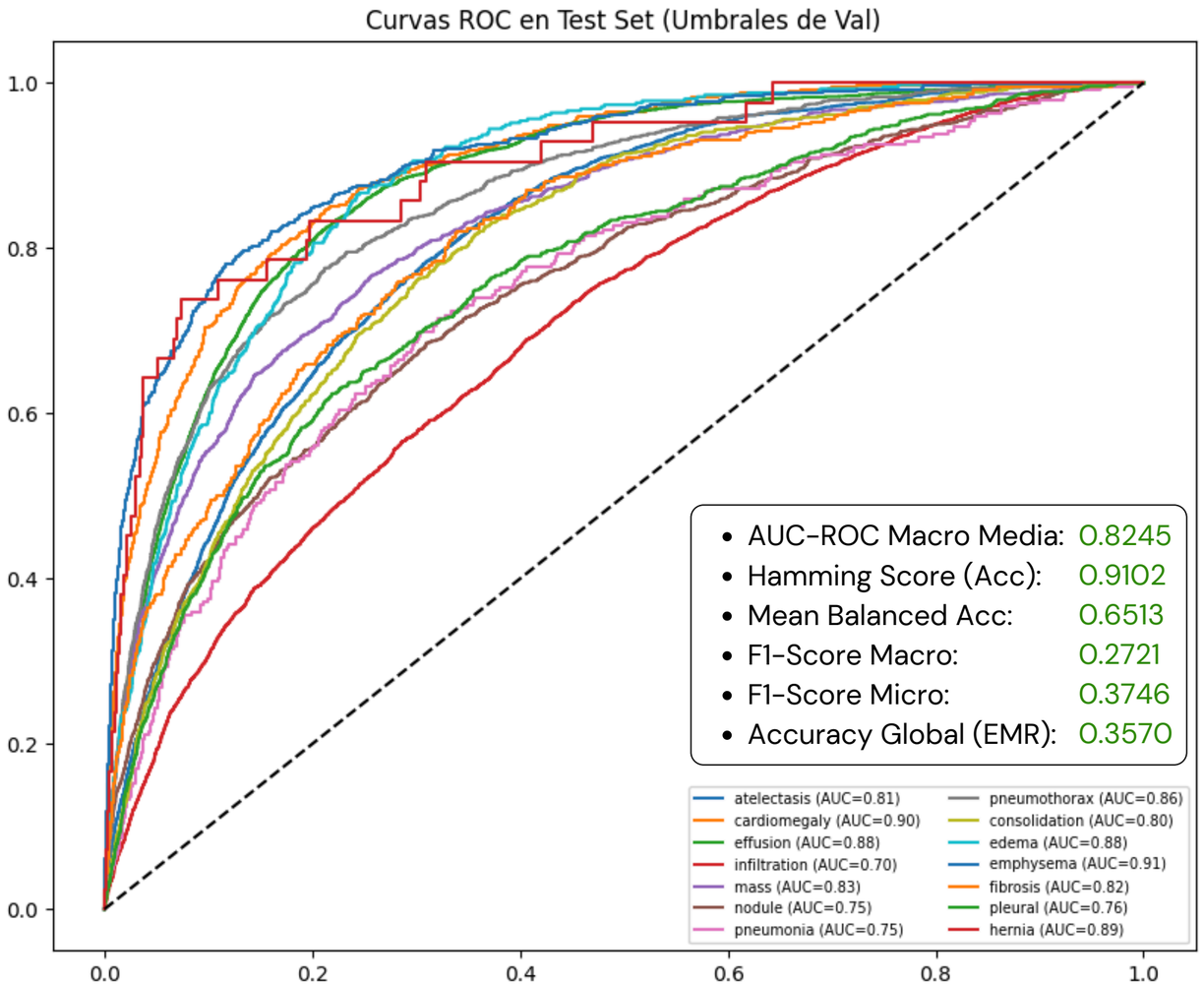
- 128x128
- 224x224

BASELINE

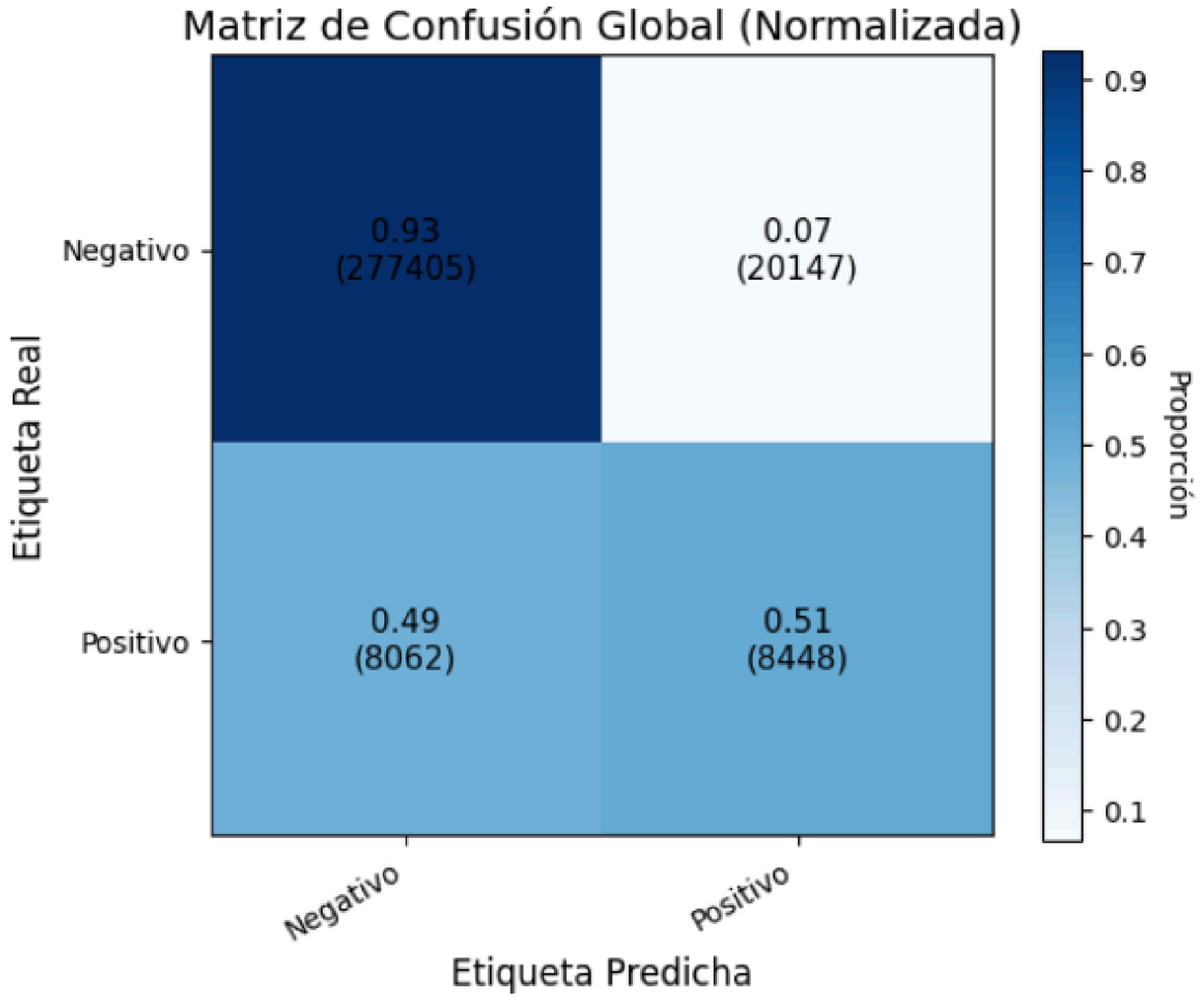
- ResNet-50 pretrained en ImageNet
- Full Fine-Tuning
- Lr 1e-4
- 50 epochs + 2 Early Stopping
- Normalización de ImageNet
- BCEWithLogitsLoss
- Umbrales por patología definidos en Validación

128x128 vs 224x224





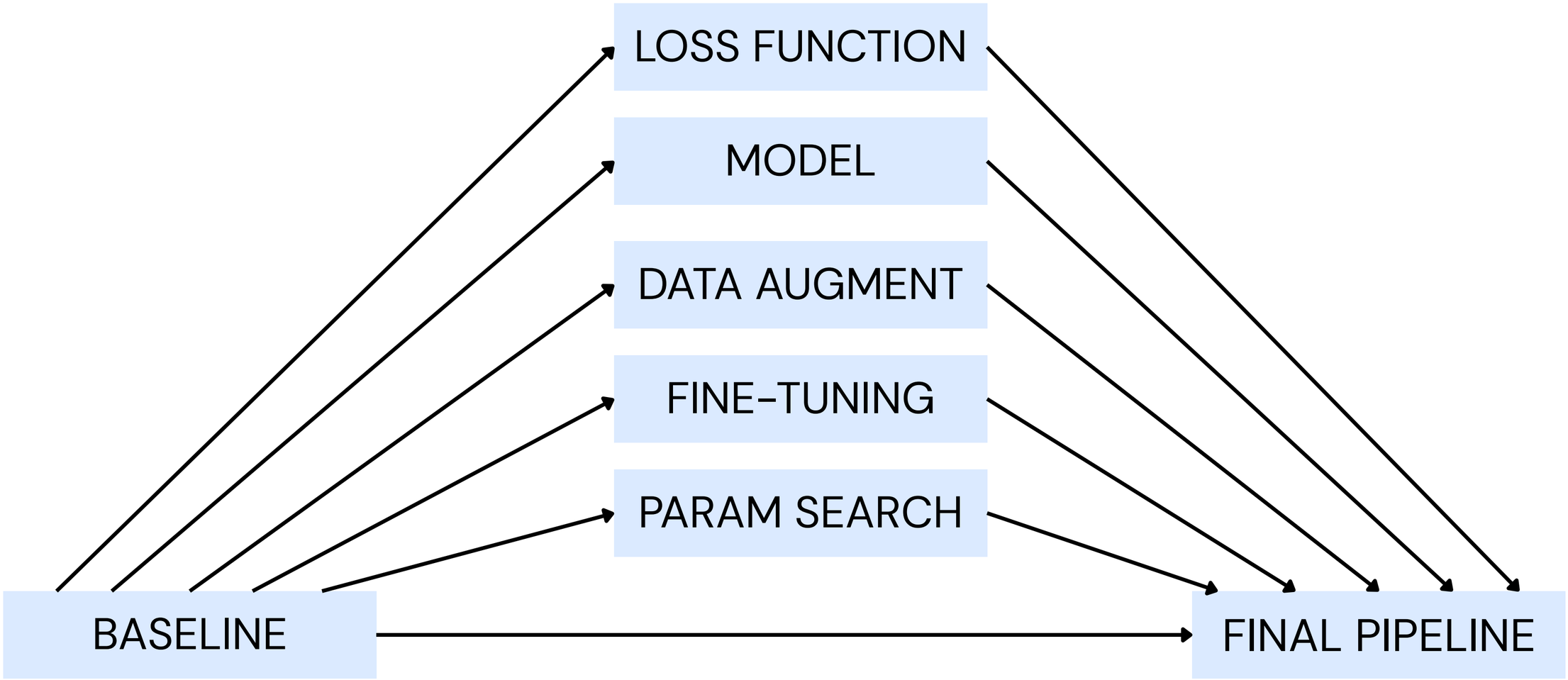
224x224


Matriz de Confusión Global (Normalizada)
Negativo
Positivo
0.93
(277405)
0.07
(20147)
0.49
(8062)
0.51
(8448)
Etiqueta Real
Negativo
Positivo
Etiqueta Predicha
Proporción
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1

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ESTUDIO DEEP LEARNING



LOSS FUNCTION

- **Binary Cross Entropy With Logits Loss:** sigmoid + BCE. Trata igual a positivos y negativos.

$$\mathcal{L}(x, y) = -\left[y \log(\sigma(x)) + (1 - y) \log(1 - \sigma(x))\right]$$

- **BCEWithLogitsLoss with weighted pos:** penaliza más si falla un positivo. No distingue ejemplos fáciles de difíciles.

$$\mathcal{L}(x, y) = -\left[w_p \cdot y \log(\sigma(x)) + (1 - y) \log(1 - \sigma(x))\right]$$

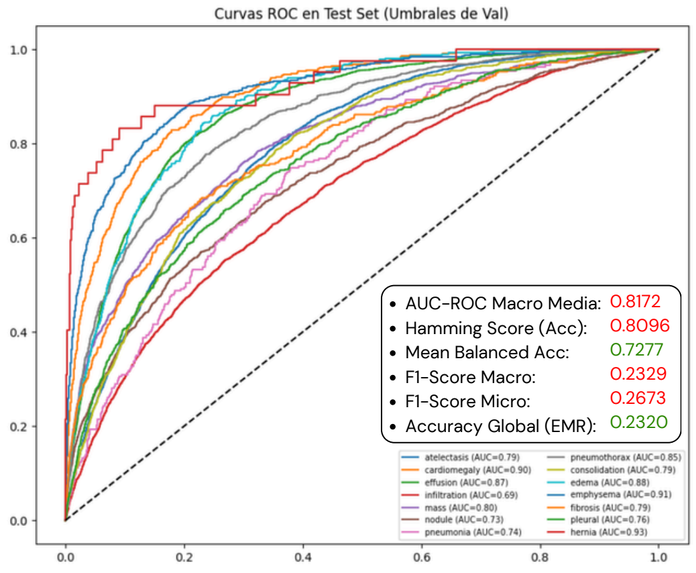
- **Focal Loss:** ignora ejemplos fáciles centrándose en los difíciles.

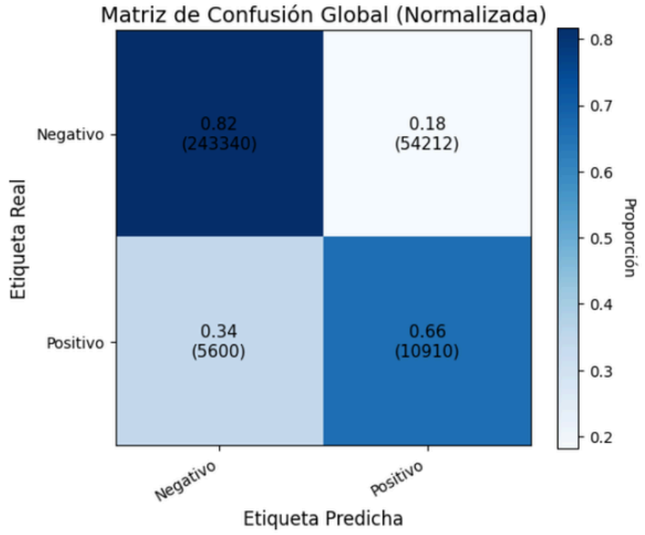
$$\mathcal{L} = \alpha(1 - p_t)^{\gamma} \cdot \mathrm{BCE}(x, y)$$

- **Asymmetric Loss:** se enfoca en ejemplos difíciles y trata distinto a positicos y negativos.

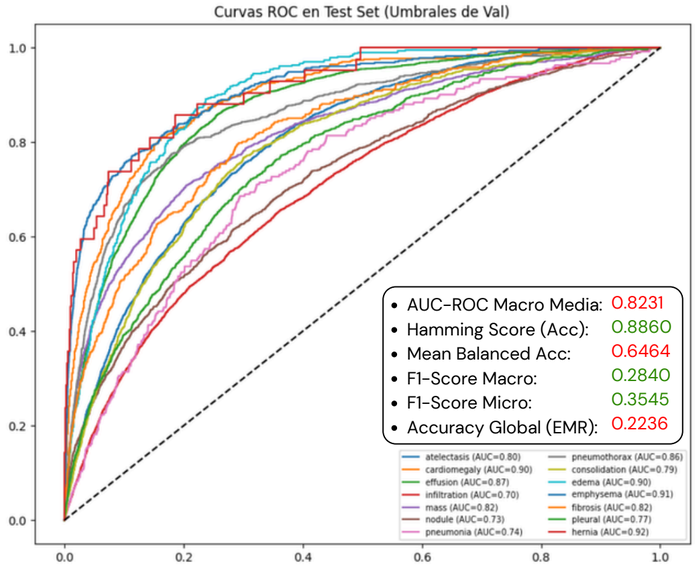
$$\mathcal{L} = -(y \log(p) + (1 - y) \log(1 - p)) \cdot (1 - p_t)^{\gamma_{pos/neg}}$$

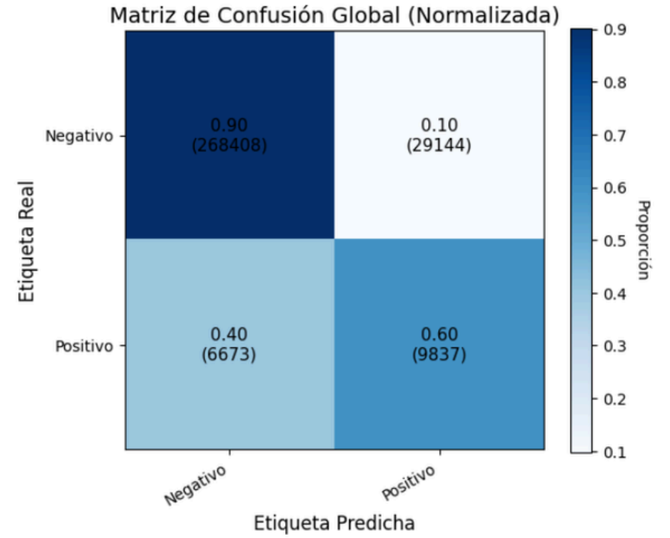
Weighted Loss



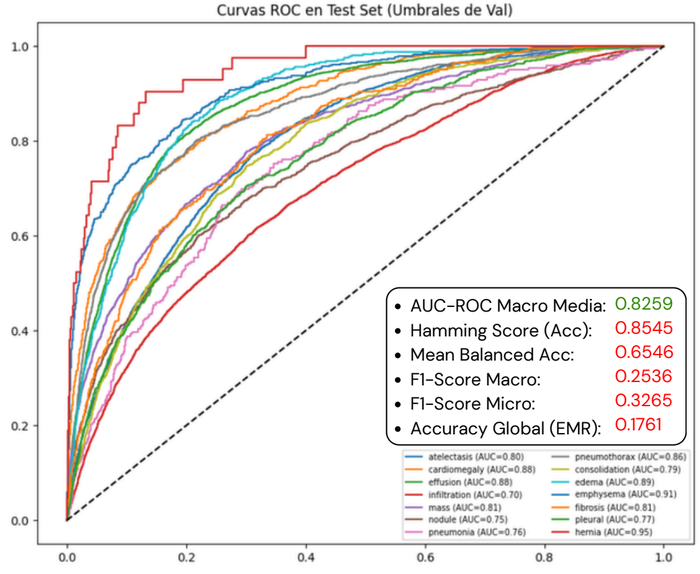


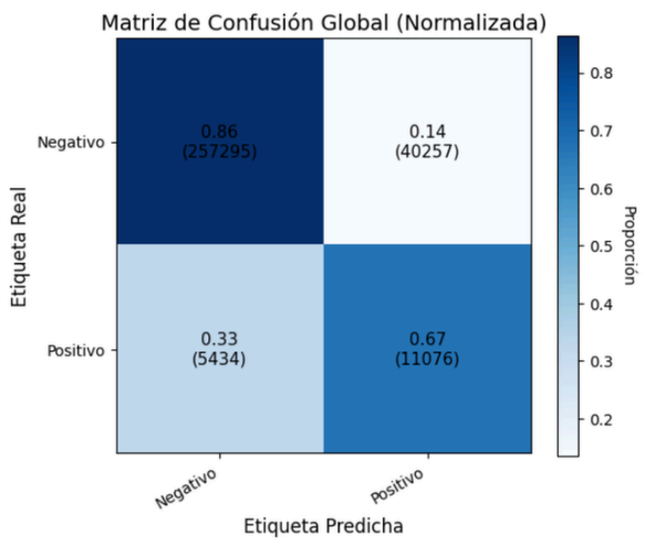
Focal Loss



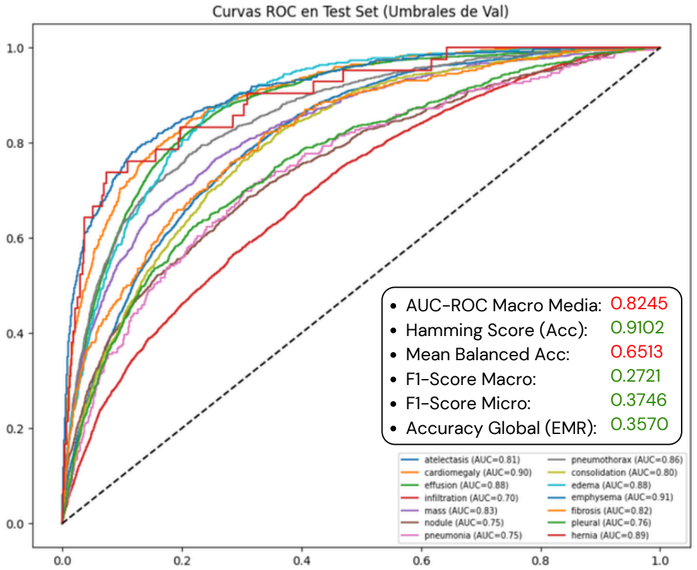


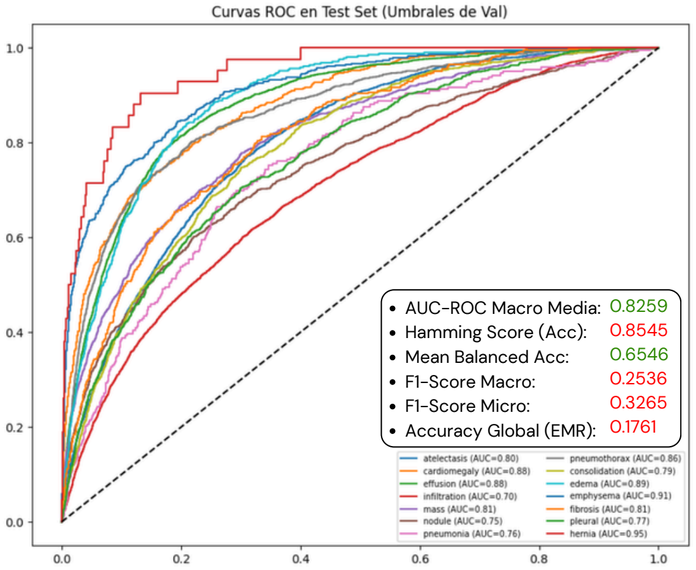
Asymmetric Loss

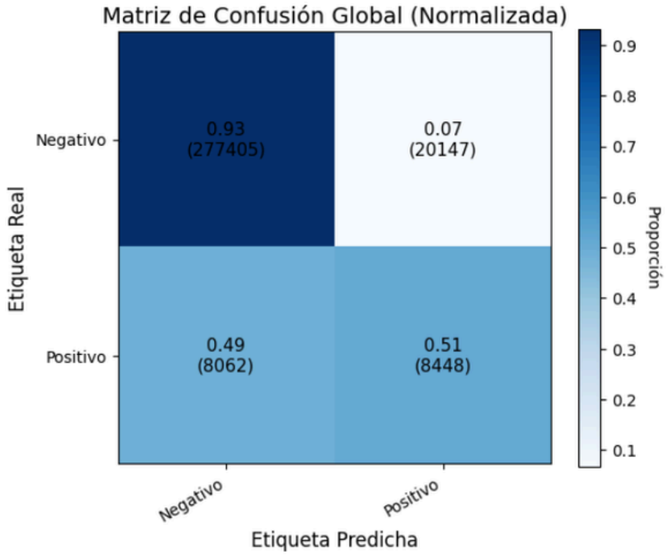


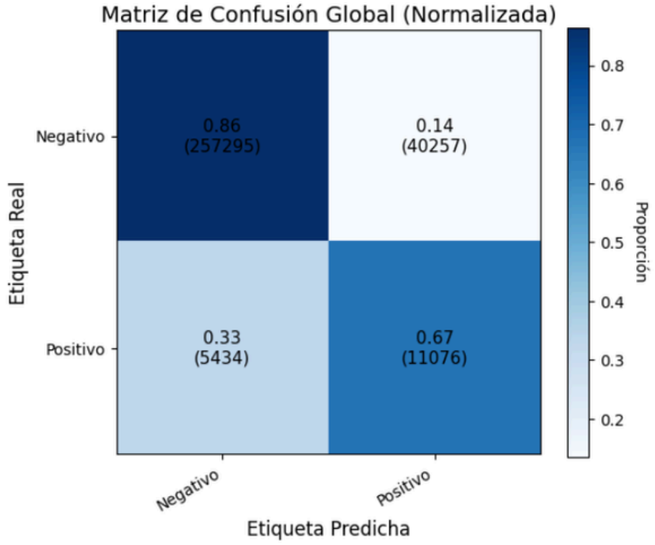


BCEWITHLOGITSLOSS VS ASYMMETRIC LOSS

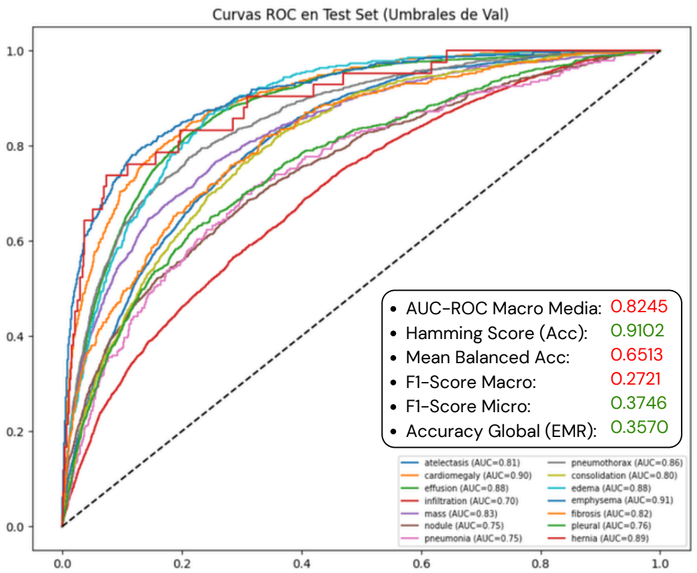


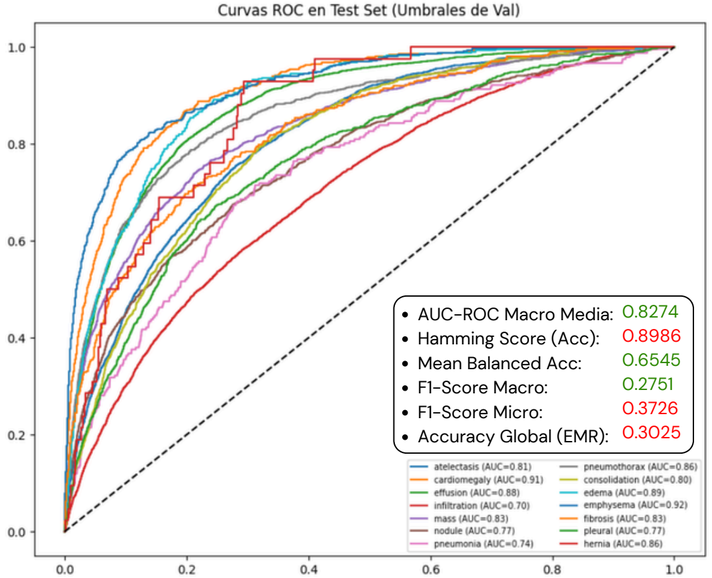


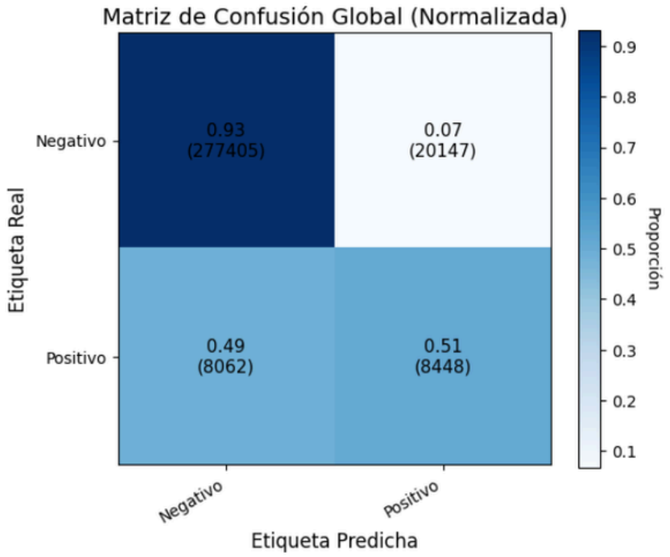


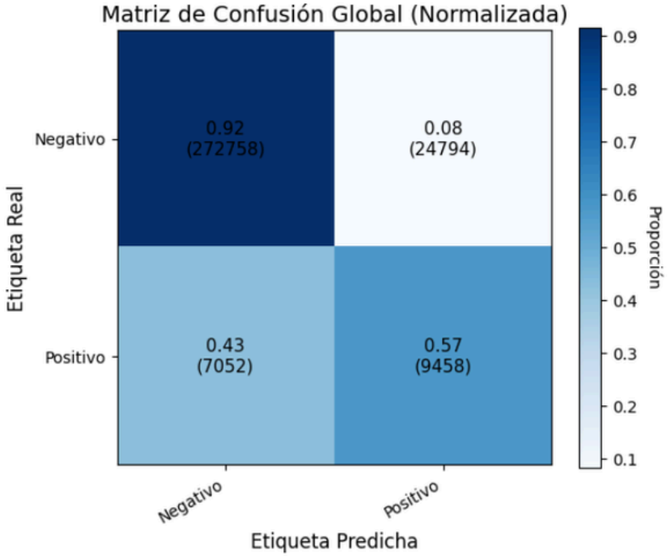


RESNET-50 VS DENSENET-121

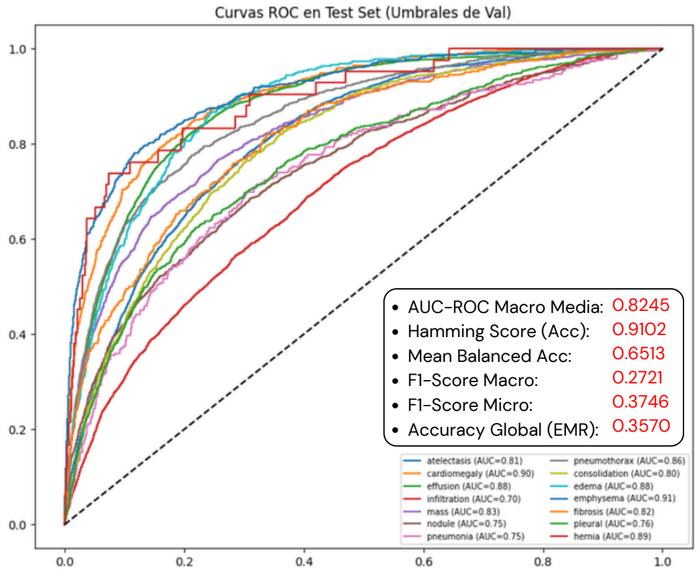


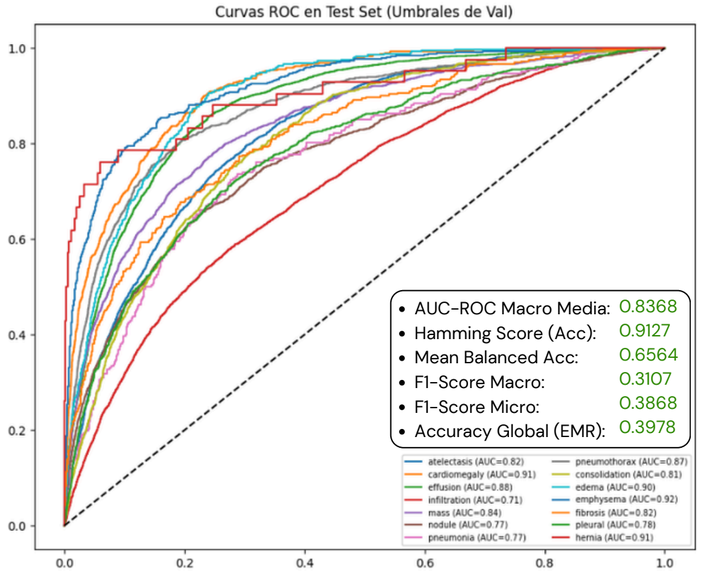


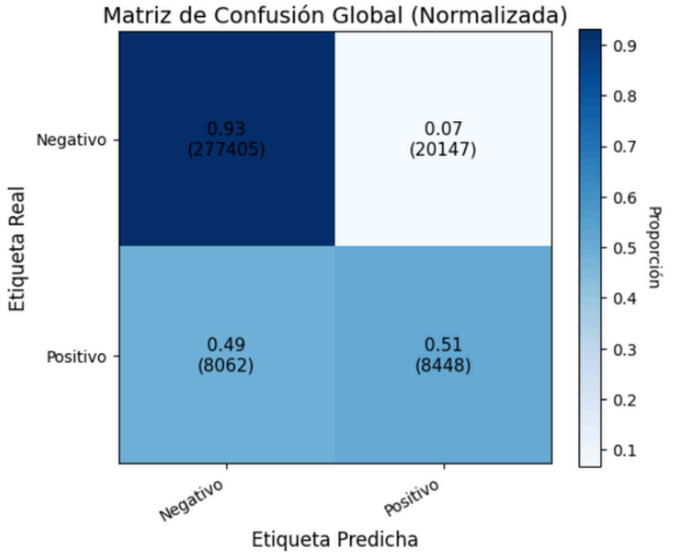


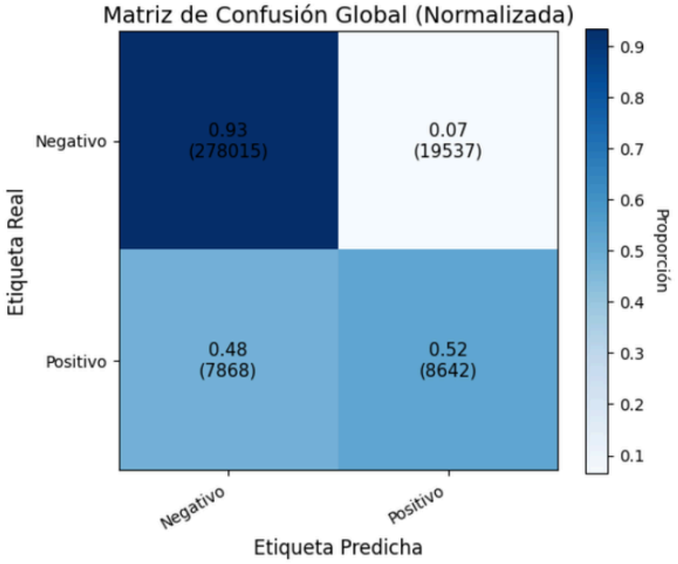


BASELINE VS DATA AUGMENTATION



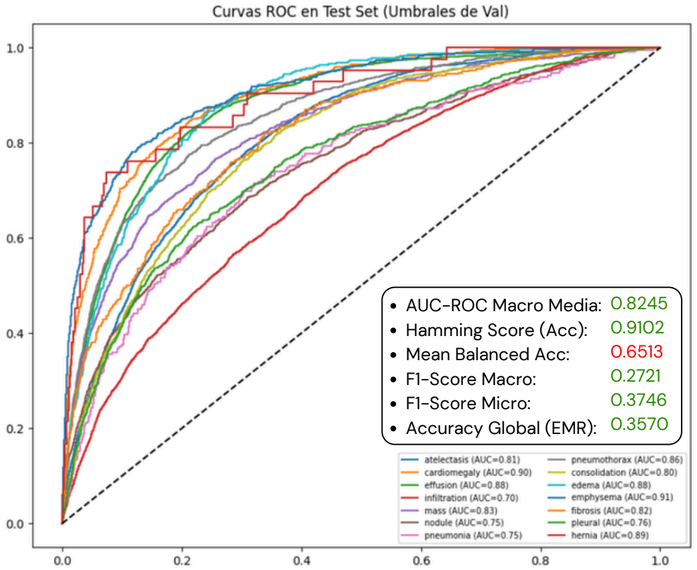


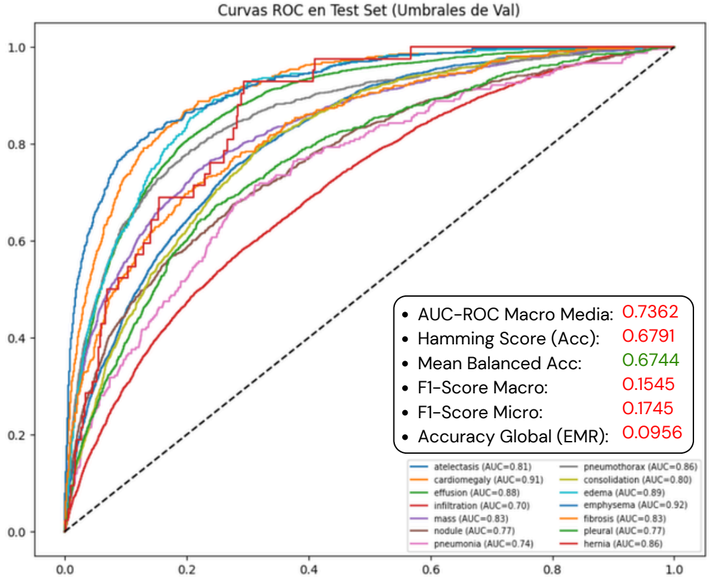


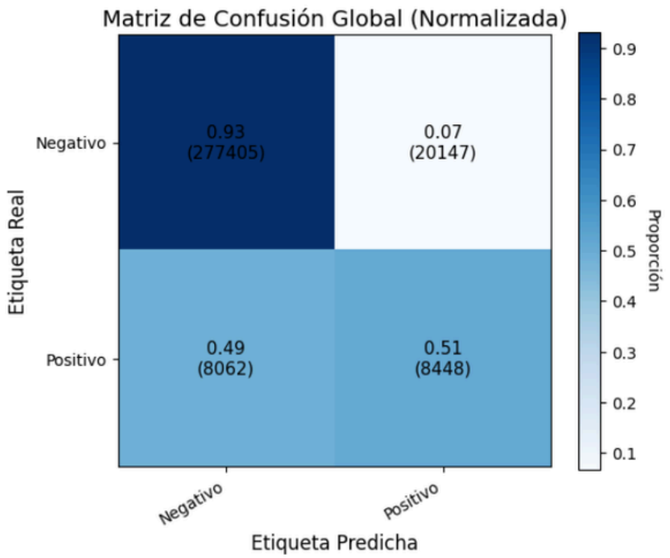


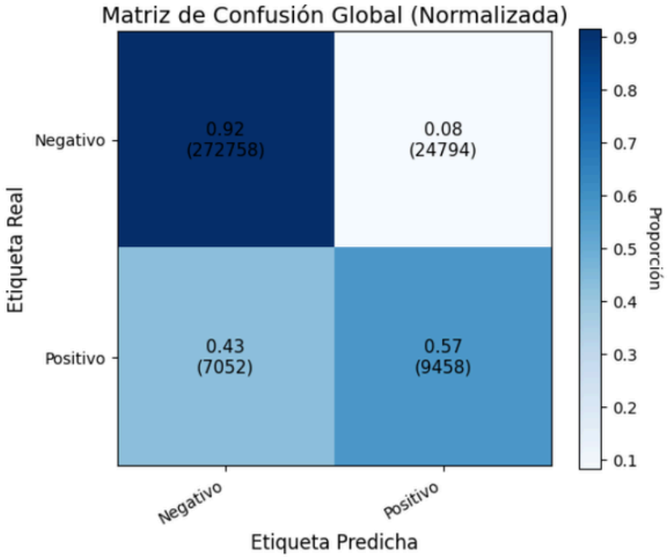
- Rotación aleatoria +−15°
- Traslación aleatoria +−10%
- Giro horizontal aleatorio

FULL F-T VS FROZEN







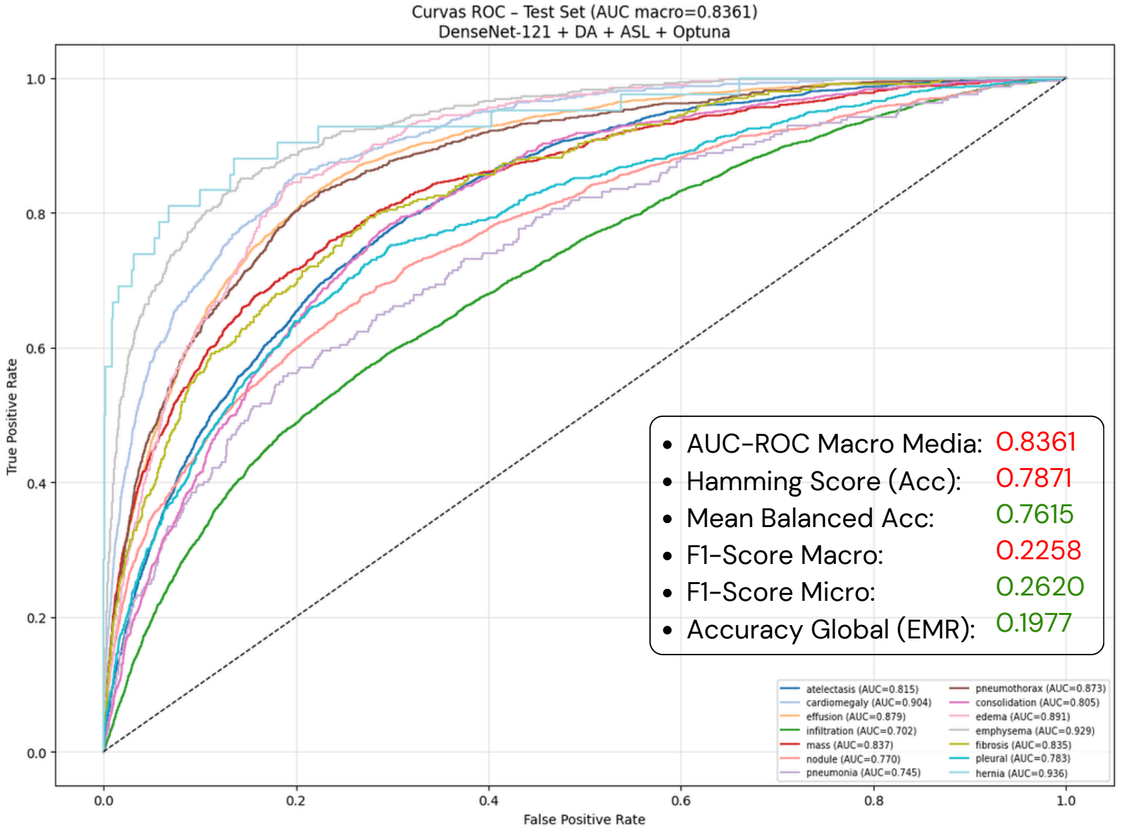


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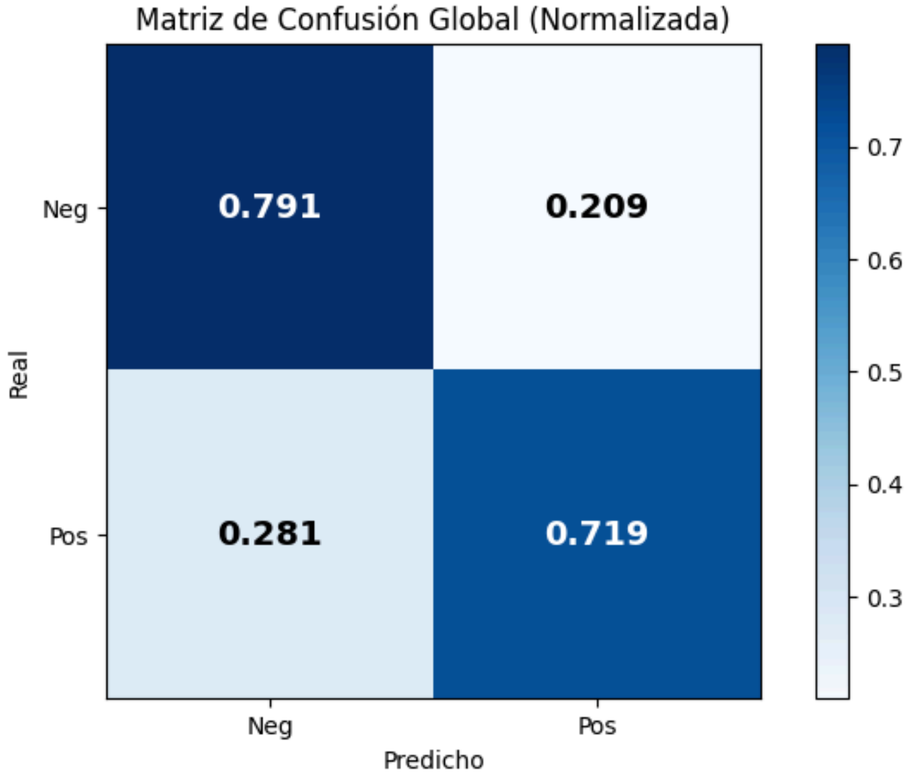
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PIPELINE FINAL

DenseNet-121 + DA + Asymmetric Loss

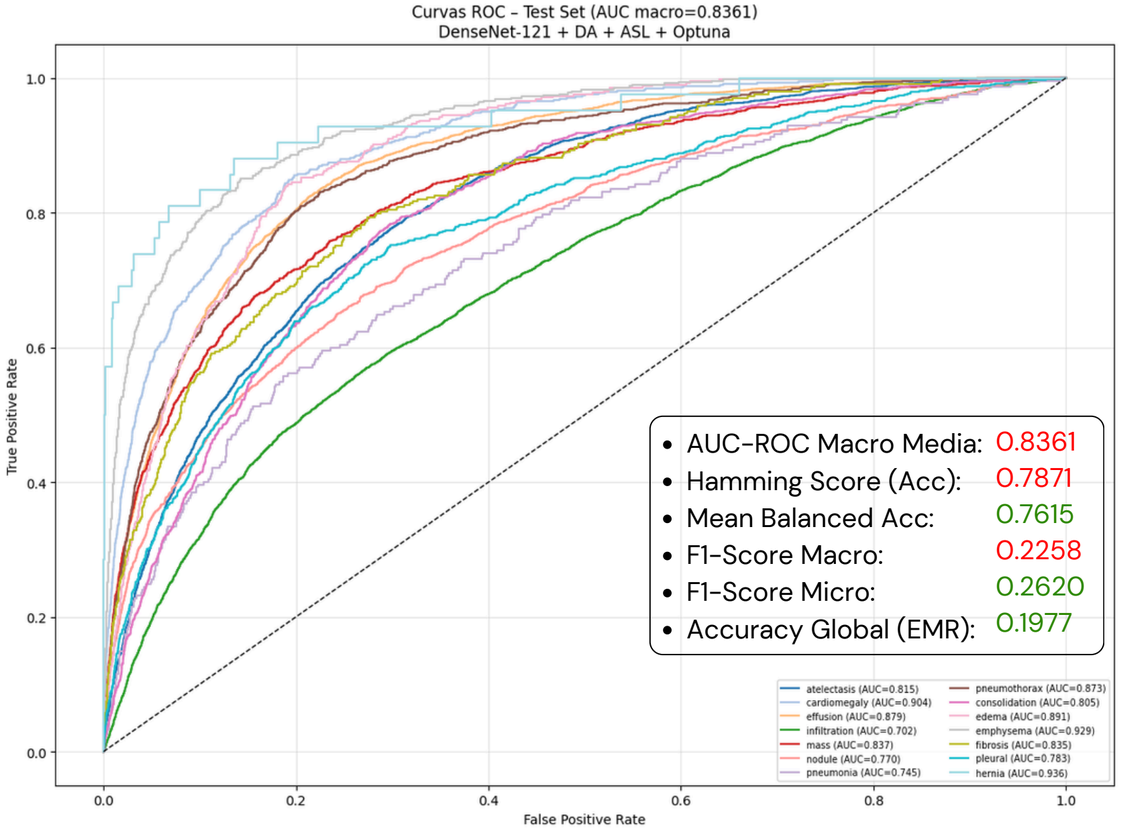


- AUC-ROC Macro Media: 0.8361 RN-DA-BCE
- Hamming Score (Acc): 0.7871 RN-DA-BCE
- Mean Balanced Acc: 0.7615
- F1-Score Macro: 0.2258 RN-DA-BCE
- F1-Score Micro: 0.2620 RN-DA-BCE
- Accuracy Global (EMR): 0.1977 RN-DA-BCE

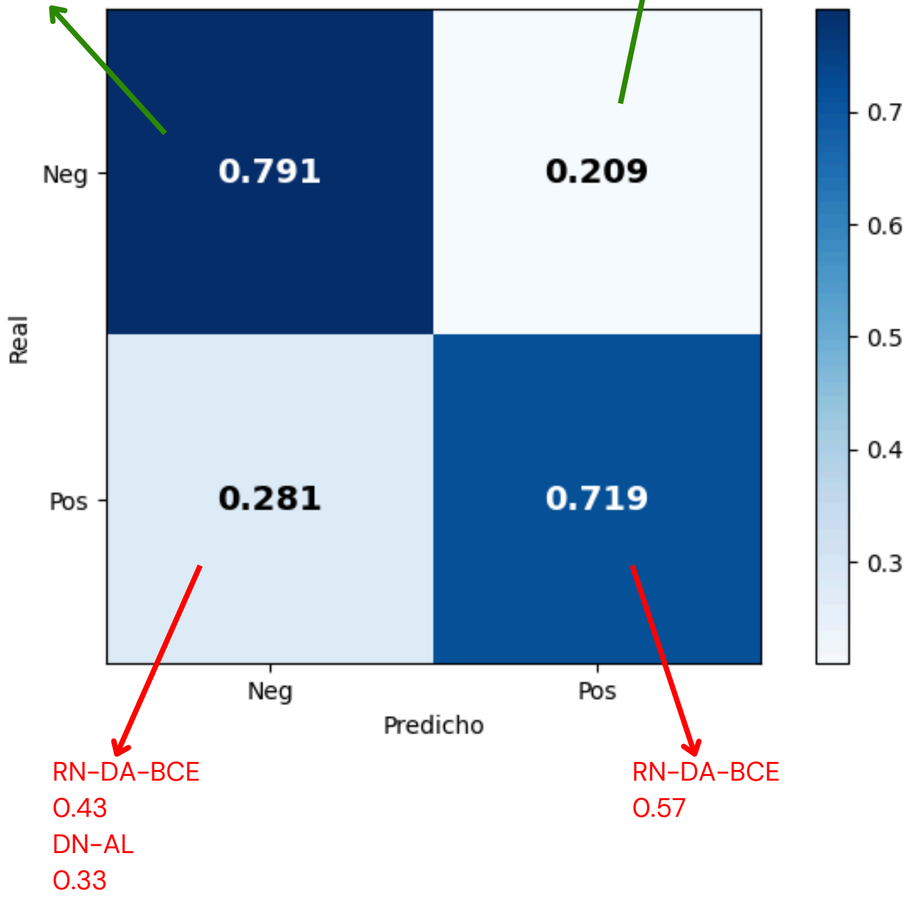


PIPELINE FINAL

DenseNet-121 + DA + Asymmetric Loss



- AUC-ROC Macro Media: 0.8361 RN-DA-BCE
- Hamming Score (Acc): 0.7871 RN-DA-BCE
- Mean Balanced Acc: 0.7615 RN-DA-BCE
- F1-Score Macro: 0.2258 RN-DA-BCE
- F1-Score Micro: 0.2620 RN-DA-BCE
- Accuracy Global (EMR): 0.1977 RN-DA-BCE



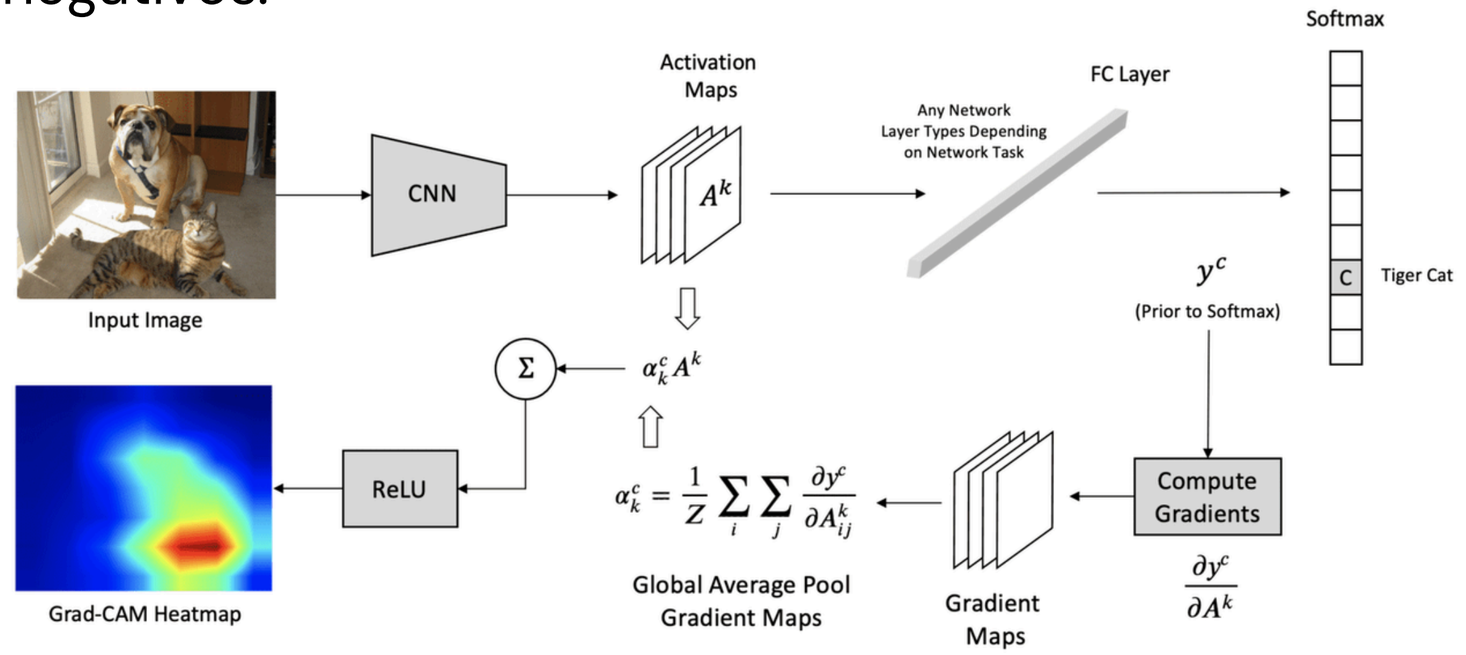
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UNA PINCELADA DE EXPLICABILIDAD (XAI)

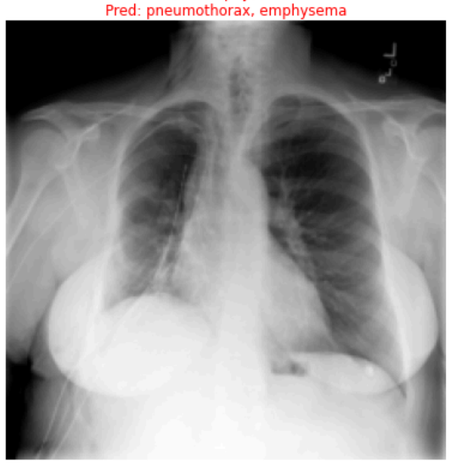
Grad-CAM: qué neuronas se activaron, dónde miraban espacialmente.

1. Forward: saca mapas de activación y probabilidad de etiqueta.
2. Backprop: guarda los gradientes, no actualiza pesos.
3. Pondera mapas de activación con los gradientes.
4. Colapsa con la suma los mapas ponderados y aplica ReLU eliminando valores negativos.

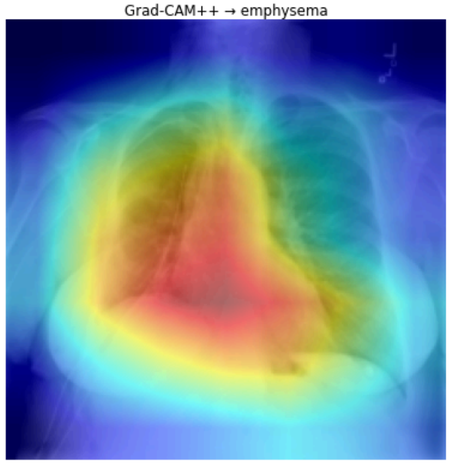


CLASES FÁCILES (ALTO AUC)

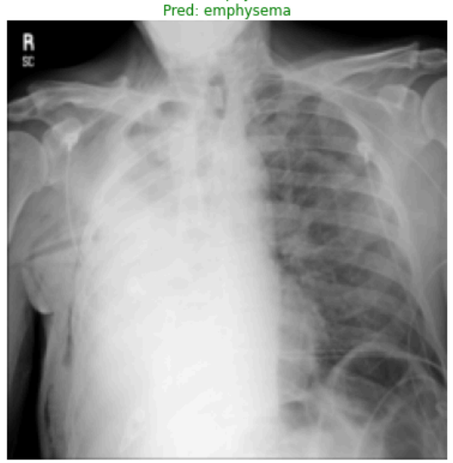
#295 | fácil·correcto
CAM: emphysema (p=0.55)
Real: emphysema
Pred: pneumothorax, emphysema



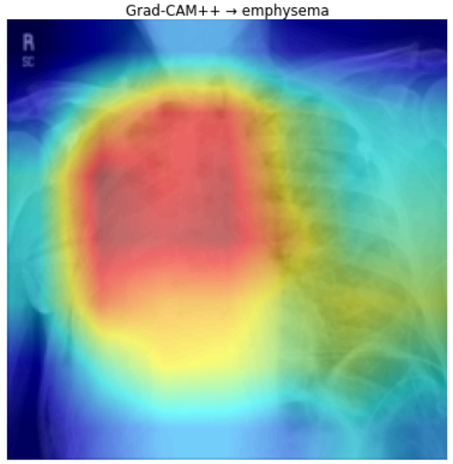
Grad-CAM++ → emphysema



#450 | fácil·correcto
CAM: emphysema (p=0.57)
Real: emphysema
Pred: emphysema



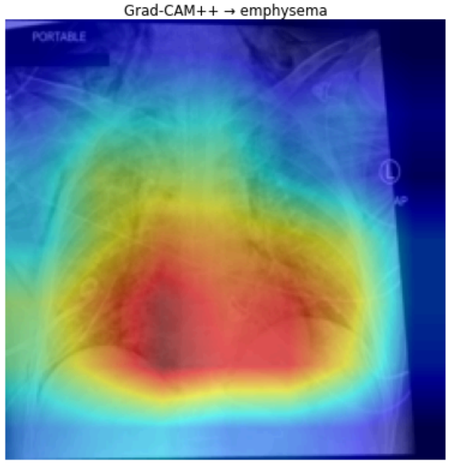
Grad-CAM++ → emphysema



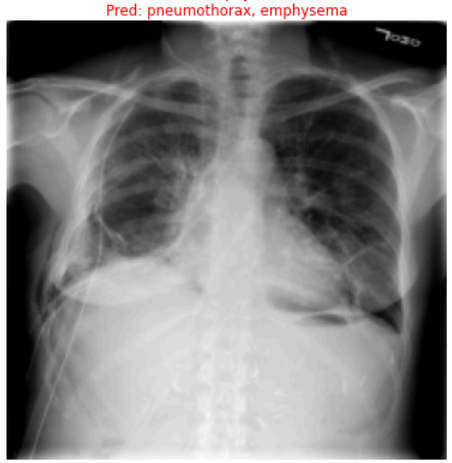
#637 | fácil·correcto
CAM: emphysema (p=0.65)
Real: emphysema
Pred: emphysema



Grad-CAM++ → emphysema



#1554 | fácil·correcto
CAM: emphysema (p=0.55)
Real: emphysema
Pred: pneumothorax, emphysema



Grad-CAM++ → emphysema

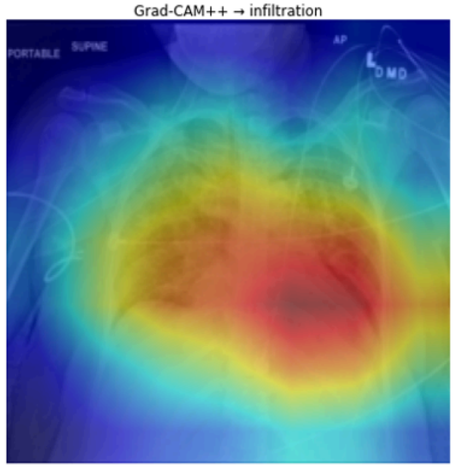


CLASES DIFÍCILES (BAJO AUC)

#17 | difícil·correcto
CAM: infiltration (p=0.56)
Real: infiltration
Pred: infiltration



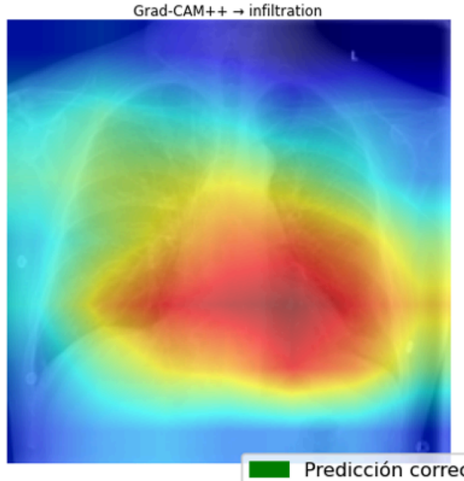
Grad-CAM++ → infiltration



#54 | difícil·correcto
CAM: infiltration (p=0.55)
Real: infiltration
Pred: infiltration



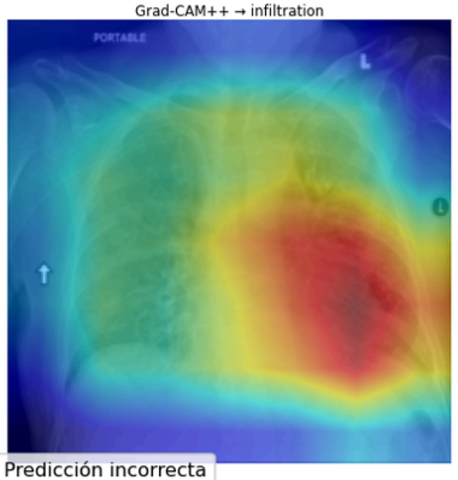
Grad-CAM++ → infiltration



#109 | difícil·correcto
CAM: infiltration (p=0.57)
Real: infiltration
Pred: effusion, infiltration



Grad-CAM++ → infiltration



#119 | difícil·correcto
CAM: infiltration (p=0.59)
Real: infiltration
Pred: infiltration



Grad-CAM++ → infiltration



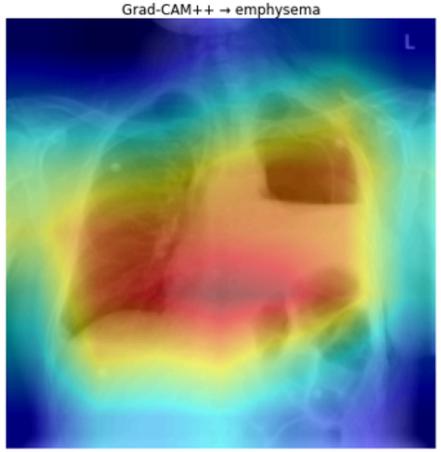
Predicción correcta Predicción incorrecta

FALSOS POSITIVOS

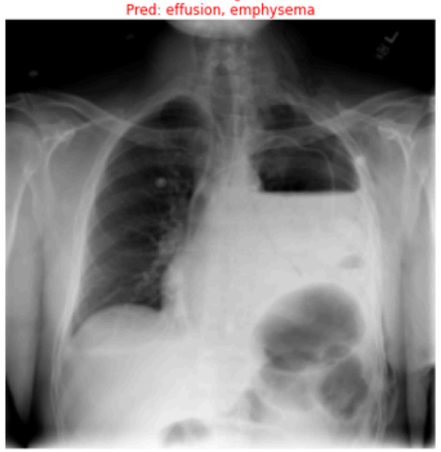
#462 | FALSO POSITIVO
CAM: emphysema (p=0.64)
Real: ninguna
Pred: effusion, pneumothorax, emphysema



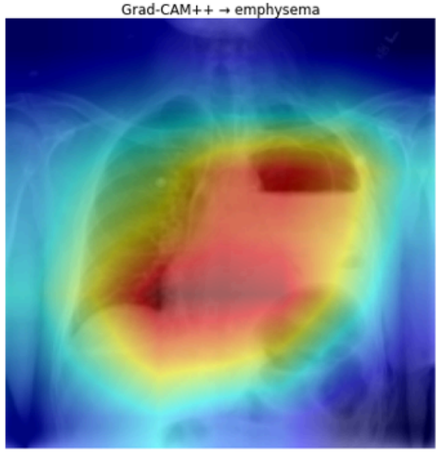
Grad-CAM++ → emphysema



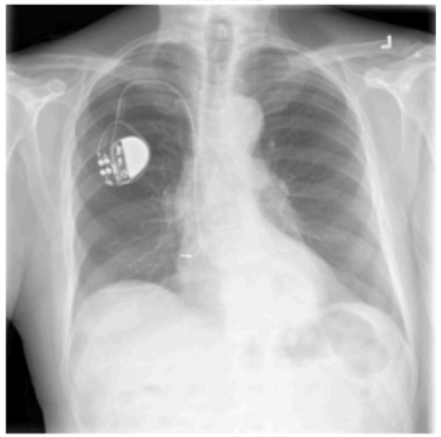
#544 | FALSO POSITIVO
CAM: emphysema (p=0.55)
Real: ninguna
Pred: effusion, emphysema



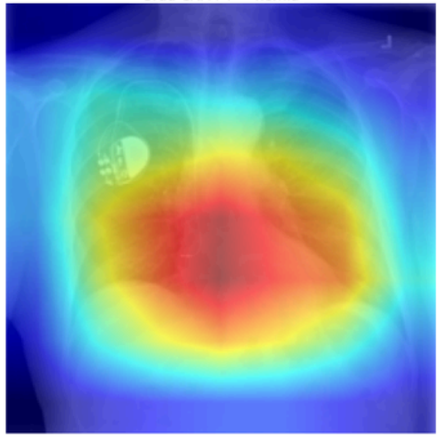
Grad-CAM++ → emphysema



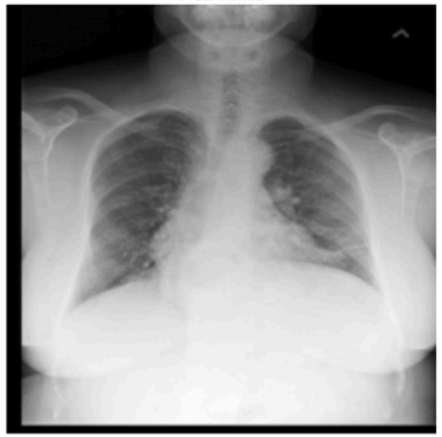
#10117 | FALSO POSITIVO
CAM: hernia (p=0.66)
Real: ninguna
Pred: hernia



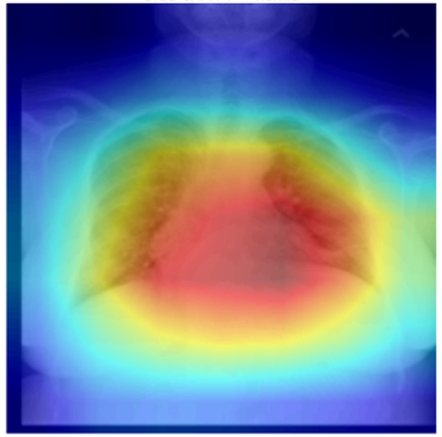
Grad-CAM++ → hernia



#10674 | FALSO POSITIVO
CAM: hernia (p=0.84)
Real: nodule
Pred: hernia

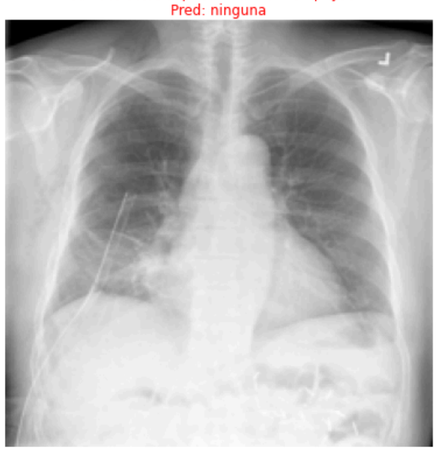


Grad-CAM++ → hernia

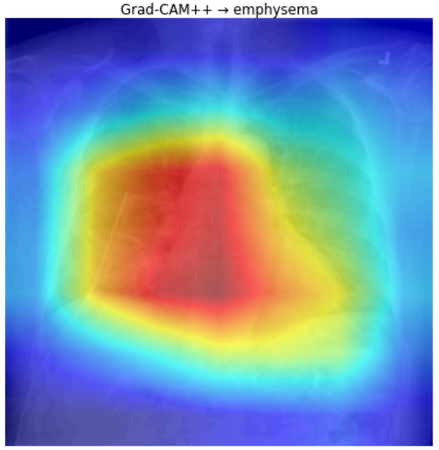


FALSOS NEGATIVOS

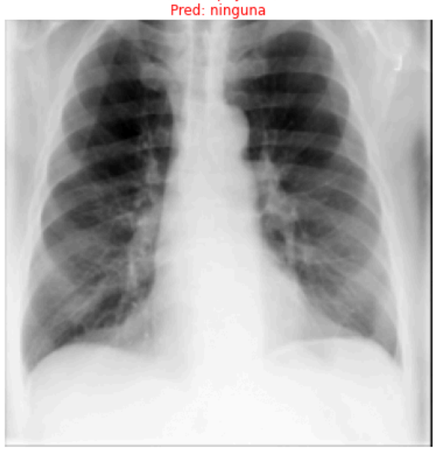
#63 | FALSO NEGATIVO
CAM: emphysema (p=0.52)
Real: atelectasis, pneumothorax, emphysema
Pred: ninguna



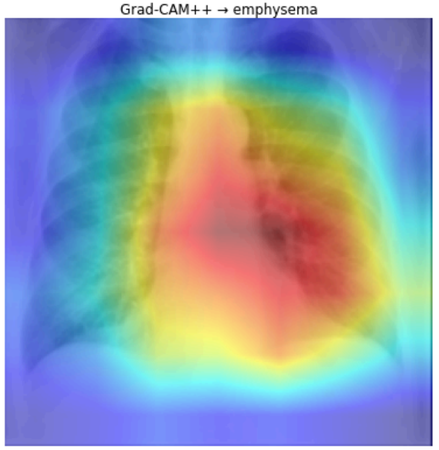
Grad-CAM++ → emphysema



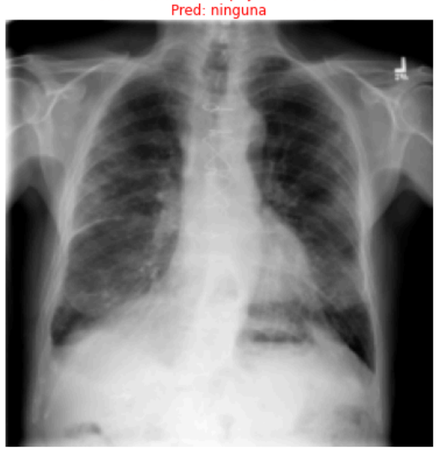
#82 | FALSO NEGATIVO
CAM: emphysema (p=0.43)
Real: emphysema
Pred: ninguna



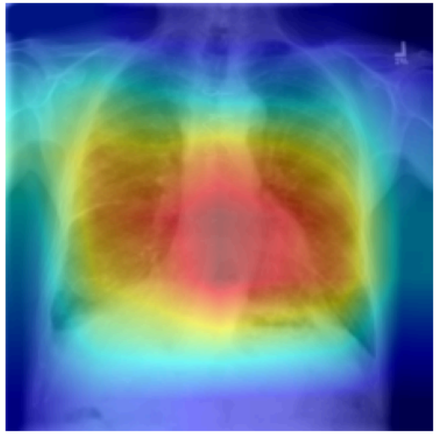
Grad-CAM++ → emphysema



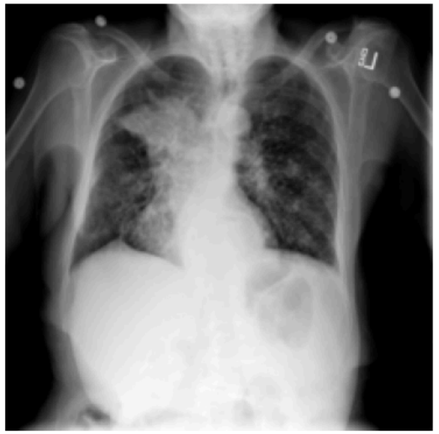
#738 | FALSO NEGATIVO
CAM: hernia (p=0.16)
Real: infiltration, emphysema, fibrosis
Pred: ninguna



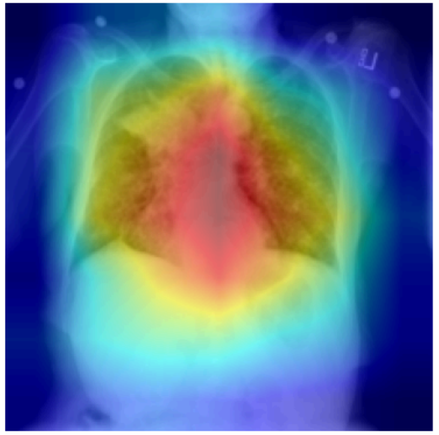
Grad-CAM++ → hernia



#1120 | FALSO NEGATIVO
CAM: hernia (p=0.43)
Real: hernia
Pred: mass

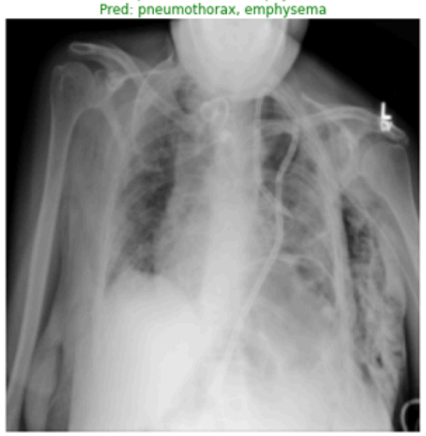


Grad-CAM++ → hernia

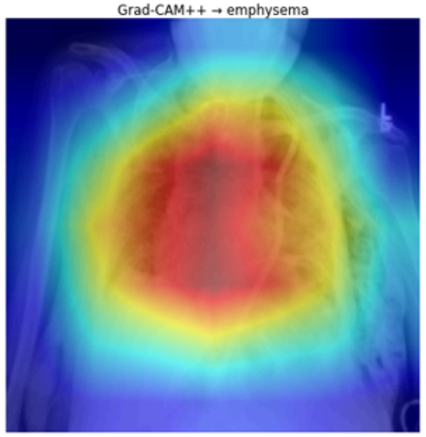


ALTA CONFIANZA

#111 | conf-alta·OK
CAM: emphysema (p=0.73)
Real: pneumothorax, emphysema
Pred: pneumothorax, emphysema



Grad-CAM++ → emphysema



#428 | conf-alta·OK
CAM: emphysema (p=0.68)
Real: consolidation, emphysema
Pred: pneumothorax, emphysema



Grad-CAM++ → emphysema



#18 | conf-alta·OK
CAM: hernia (p=0.76)
Real: hernia
Pred: hernia



Grad-CAM++ → hernia



#1595 | conf-alta·OK
CAM: hernia (p=0.90)
Real: atelectasis, hernia
Pred: hernia



Grad-CAM++ → hernia



ÍNDICE

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2. EDA
3. Métricas
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5. Estudio Deep Learning
6. Pipeline Final
7. Una pincelada de explicabilidad (XAI)
8. **Referencias**

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GRACIAS